

Conditional Statements

If... then...

Given statements P and Q , we can form a new statement “If P , then Q ”.

- ▶ If (it is raining), then (it is cloudy).
- ▶ If (a number is divisible by 6), then (it is even).

These are called **conditional statements**.

Symbolically

$$P \implies Q$$

.

If... then..

Truth of the first statement forces truth of the second statement.

$P \Rightarrow Q$ TRUE
 FALSE

P	Q
T	T
T	F
F	T
F	F

P	Q
T	T
T	F
F	T
F	F

Alternatives

- ▶ If P , then Q
- ▶ Q if P
- ▶ Q whenever P
- ▶ Q , provided that P
- ▶ Whenever P , then also Q
- ▶ P is a sufficient condition for Q

more alternatives

- ▶ For Q , it is sufficient that P
- ▶ Q is a necessary condition for P
- ▶ For P , it is necessary that Q
- ▶ P only if Q

Example

A matrix is invertible provided that its determinant is non-zero.

P $\det \neq 0$
Q invertible

$P \Rightarrow Q$

Example

An integer is divisible by 8 only if it is divisible by 4.

Example

Being a native born citizen over the age of 35 is sufficient to be eligible to be elected president of the US.

P

Q

R

native born

≥ 35

eligible

$P \text{ and } Q \Rightarrow R$

Example

Being over the age of 35 is necessary to be elected president of the US.